

```

        self.myRow = random.randrange(self.nRows) # chooses a random row
        self.myCol = random.randrange(self.nCols) # chooses a random col
        self.mySpeedX = random.randrange(-maxSpeed, maxSpeed + 1) # chooses an X
speed
        self.mySpeedY = random.randrange(-maxSpeed, maxSpeed + 1) # chooses a Y
speed
        # Set other instance variables like health, power, etc.

    def move(self):
        self.myRow = (self.myRow + self.mySpeedY) % self.nRows
        self.myCol = (self.myCol + self.mySpeedX) % self.nCols

```

---

*Listing 4-12: A `Monster` class that can be used to instantiate many `Monster`s*

With this `Monster` class, we can create a list of `Monster` objects like this:

```

N_MONSTERS = 20
N_ROWS = 100    # could be any size
N_COLS = 200    # could be any size
MAX_SPEED = 4
monsterList = [] # start with an empty list
for i in range(N_MONSTERS):
    oMonster = Monster(N_ROWS, N_COLS, MAX_SPEED) # create a Monster
    monsterList.append(oMonster) # add the Monster to our list

```

---

This loop will instantiate 20 `Monster`s, and each will know its own starting location in the grid and its individual speed. Once you have a list of objects, later in the program when you want each object to do the same action, you can write a simple loop where you call the same method of each object in the list:

```

for objectVariable in objectVariablesList:
    objectVariable.someMethod()

```

---

For example, if we want each of our `Monster` objects to move, we could use a loop like this:

```

for oMonster in monsterList:
    oMonster.move()

```

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